Category-Aware Explainable Machine Learning for Urban Service Demand Forecasting in Smart Cities

Tran Dai Phong Lam[1*], Thu Le[1], Nguyen Quoc Hung[1], Nguyen Trung Trinh[1]

[1*]FPT University, Ho Chi Minh City, Vietnam.

*Corresponding author(s). E-mail(s): phonglam2599@gmail.com;
Contributing authors: thulvm@fpt.edu.vn; hungtvt2222@gmail.com; trinhnguyen112355@gmail.com;

Abstract

Urban service platforms such as 311 systems create large-scale traces of reported municipal demand, but forecasting operationally relevant reported-demand surges remains difficult because requests are spatially uneven, service categories are heterogeneous, temporal patterns are non-stationary, and reporting behavior affects observed counts. This paper formulates reported urban service-demand forecasting as a spatiotemporal early-warning task at the neighborhood-week-complaint-category level. Using New York City as a case study, we integrate more than 30.9 million geocoded 311 requests with historical demand dynamics, calendar timing, NOAA weather exposure, OpenStreetMap point-of-interest context, and PLUTO land-use and built-environment descriptors. The target is a next-week abnormal reported-demand event, defined from shifted historical rolling statistics so that forecasting features do not leak future information. Models are evaluated with a target-week chronological protocol: target weeks in 2015-2022 for training, 2023 target weeks for validation, and 2024-2025 target weeks for held-out future testing; validation and test positive shares are approximately 12.7%. The validation-selected category-aware decision strategy averages LightGBM and XGBoost probability scores and applies complaint-category-specific thresholds selected only on the validation period. Under this target-week chronological evaluation, it achieves future-test F1 = 0.3839, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, and ROC-AUC = 0.7720. SHAP analysis of the LightGBM component shows that historical temporal demand dynamics dominate prediction, while built-environment, POI, weather, service-category, and calendar features explain additional variation not captured by recent demand history. These results position the model as a decision-support tool: it identifies neighborhoods and service categories with elevated risk of abnormal reported demand, while recognizing that 311 data capture reported service demand rather than all underlying urban conditions.

Keywords: urban computing, smart cities, 311 service requests, service-demand forecasting, explainable machine learning, SHAP, spatiotemporal modeling

1 Introduction

Smart-city service systems increasingly rely on administrative data streams to understand where and when public needs emerge. Among these streams, 311 requests are valuable because they record resident-facing reports across municipal service domains. Prior work shows that 311 data contain spatial signatures of urban conditions [1], while also reflecting uneven participation and reporting behavior [2]. Thus, 311 data can support city operations and research, but should not be treated as a complete census of all urban problems.

We use NYC 311 requests as a large-scale case study for forecasting reported urban service demand. Rather than predicting raw complaint counts, we formulate an early-warning task: for each neighborhood, week, and complaint category, estimate whether the following week will exhibit an abnormally high reported-demand event relative to local history. This formulation is closer to operational use, where analysts often need near-term, category-specific warnings rather than a single citywide count.

We build a multi-source and explainable pipeline with leakage-safe target and temporal-feature construction over a dense NTA-week-complaint-category panel. Historical 311 demand is combined with calendar timing, weekly NOAA weather summaries, OSM POI functional context, and PLUTO land-use and built-environment descriptors. Because the target is a next-week abnormal reported-demand event, all lag and rolling-window features are shifted so that only pre-prediction information is used.

The empirical evaluation follows a target-week chronological design. Models are trained using target weeks in 2015-2022, tuned using 2023 target weeks, and evaluated on a held-out future test period covering 2024-2025 target weeks. This split is central because random splits would mix future and past observations from the same neighborhood-category series and overstate forecasting validity. Since next-week abnormal reported-demand events are relatively rare, the evaluation emphasizes F1, precision, recall, PR-AUC, ROC-AUC, balanced accuracy, and confusion matrices rather than raw accuracy.

The validation-selected category-aware strategy averages tuned LightGBM and XGBoost scores and applies complaint-category-specific thresholds selected on validation. It is a decision-support layer, not an automated allocation system: it ranks potential abnormal reported-demand events while making false positives and false negatives visible. We explain a tuned LightGBM component using SHAP to identify whether predictions are driven by historical dynamics, weather, service category, POI context, land use, or calendar effects.

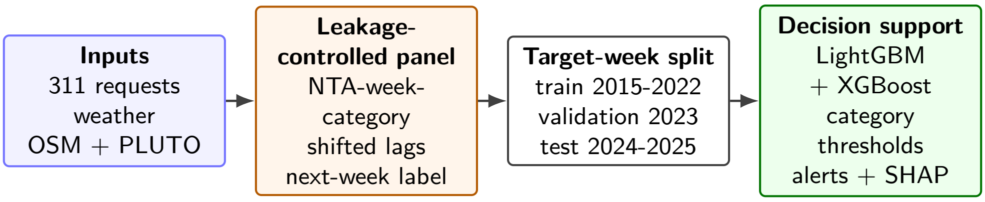

Fig. 1 End-to-end leakage-controlled temporal workflow. Feature week t predicts target week $t+1$; chronological periods are split by target week before fitting; thresholds are selected only on validation; and SHAP reports normalized summed mean absolute contribution patterns for reported demand rather than causal effects.

We organize the evaluation around three research questions: (RQ1) can a multi-source panel with leakage-safe target and temporal-feature construction forecast next-week abnormal reported demand under a future chronological test; (RQ2) how do category-specific thresholds alter precision-recall and alert-rate trade-offs across service categories; and (RQ3) which temporal and urban-context feature groups drive the predictions? The contributions are: (1) a category-aware early-warning formulation at the service-domain level; (2) a large NYC panel built from 30,940,129 NTA-matched 311 requests and multi-source urban context; (3) a target-week chronological evaluation with validation-only threshold selection, baselines, and feature ablations; and (4) SHAP explanations that distinguish predictive associations from causal claims. Figure 1 summarizes this workflow.

2 Related Work

2.1 Urban computing and citizen-generated service data

Urban computing integrates sensing, administrative, and geographic data to understand city dynamics and support decision making [3]. In this setting, 311 service requests link residents, reported concerns, and municipal response systems. Prior work shows that 311 request structure can serve as a signature of urban location [1], but citizen-generated data also reflect awareness, trust, access, reporting propensity, and governance processes [2]. These findings motivate the central caution in this paper: we forecast abnormal reported service demand, not the full set of underlying urban problems. Domain studies such as NYC flooding analysis through 311 complaints [4] further show why weather and service reports can be linked, while our contribution is a citywide multi-category next-week warning task.

2.2 Urban context from POIs and land use

Neighborhood function shapes reported service demand. OSM is a major volunteered geographic information source [5], and OSM POIs have been used to model urban change [6]. Land-use and built-environment data add stable neighborhood context, so we use aggregate POI, diversity, and PLUTO features as contextual signals rather than as exact measurements of demand.

2.3 Imbalanced forecasting, gradient boosting, and explainability

The target in this paper is a next-week abnormal reported-demand event: a supervised and operationally defined next-week label. Since positive events are much less frequent than normal observations, PR-AUC and F1 are more informative than accuracy alone. Saito and Rehmsmeier [7] show why precision-recall evaluation is especially useful for imbalanced classification.

Gradient-boosted tree methods are strong tabular-learning baselines and are well suited to nonlinear interactions between temporal and contextual features. The modeling stack uses XGBoost [8] and LightGBM [9] as the main tuned models. Because operational use of such models requires interpretability, we apply SHAP [10] and TreeSHAP [11] to explain a tuned LightGBM component. We follow XAI cautions [12]: explanation values describe model behavior and should not be interpreted as causal effects.

3 Data and Task Formulation

3.1 Data sources and analytical unit

The project integrates five evidence sources from official public data portals: NYC Open Data 311 Service Requests from 2015-2025 spatially joined to 2020 NTAs; nine stable complaint-category groups; NOAA Global Historical Climatology Network Daily weather aggregated to weekly exposure; OpenStreetMap POIs aggregated to NTA-level counts, density, entropy, and mixed-use indicators; and NYC Planning PLUTO/MapPLUTO land-use and built-environment summaries. Weekly 311 counts use the Created Date field and Monday week starts. Weather comes from NOAA GHCN-Daily Central Park station `GHCND:USW00094728` and is aggregated to weekly city-level exposure. NTA assignment uses NYC Planning 2020 NTA boundaries. OSM POI and PLUTO/MapPLUTO variables are generated from fixed extraction snapshots downloaded on 2026-06-25, so they are treated as static context descriptors rather than dynamic covariates.

The resulting analytical unit is

$$(n, t, c),$$

where n denotes an NTA, t a Monday week start date, and c a complaint category. The dense panel contains 262 NTAs, 575 weekly periods, and nine complaint categories, yielding 1,355,850 rows. After target construction, 1,334,628 rows have a valid next-week label. Relative to the full dense panel, 30,654 rows are excluded due to warm-up/history requirements and unavailable next-week targets, leaving 1,325,196 model-ready rows. Table 1 summarizes the most important construction facts.

3.2 Complaint-category mapping

NYC 311 complaint types are highly granular and can change over time. Modeling each raw type separately would create sparse series, while collapsing all requests would

Table 1 Dataset construction summary.

Item	Value	Item	Value
NTA-matched 311	30,940,129	Match rate	99.9777%
Unit	NTA-week-category	Panel rows	1,355,850
Valid-label rows	1,334,628	Model-ready rows	1,325,196
Excluded rows	30,654	NTAs	262
Weeks	575	Categories	9
Coverage	2014-12-29-2025-12-29	NOAA days	4,018
OSM POIs	77,083	Output columns	252
Predefined schema	215 inputs	Final feature subset	213 inputs

erase service meaning. We therefore use stable service categories: noise, housing, water and sewer, public safety, parking and traffic, environmental issues, infrastructure, sanitation, and a residual other category retained to avoid false precision. Representative raw types include Noise–Residential and Noise–Street/Sidewalk for noise, Heat/Hot Water and Plumbing for housing, Illegal Parking and Blocked Driveway for parking/traffic, Street Condition and Sidewalk Condition for infrastructure, and Missed Collection or Dirty Condition for sanitation. In this paper, category-aware refers to modeling reported demand at this stable service-category level and applying category-specific alert thresholds, rather than using raw complaint types or a single citywide alert rule.

3.3 Next-week abnormal reported-demand target

For each neighborhood n, week t, and complaint category c, the feature vector is

$$x_{n,t,c} = [\text{historical demand}, \text{calendar}, \text{weather}, \text{OSM POI context}, \text{PLUTO context}].$$

Here t is the feature week: all predictors are available after that week closes. The target week is $t+1$, seven days after the feature-week start date. The supervised target indicates whether this following week has abnormally high reported service demand:

$$y_{n,t,c} = \mathbf{1}\{\text{count}_{n,t+1,c} > \mu^{8w}_{n,t,c} + 1.5\sigma^{8w}_{n,t,c}\},$$

where $\mu^{8w}_{n,t,c}$ and $\sigma^{8w}_{n,t,c}$ are computed from a shifted historical eight-week window. The shift ensures that rolling statistics and lag features use only prior weeks available before the prediction week. The target focuses on future reported-demand events relative to each local historical baseline rather than absolute volume alone.

4 Methodology

4.1 Feature groups

The model-ready configuration contains a predefined 215-input feature schema specified before model fitting; it is not selected using validation or test labels. The final tuned comparison uses a 213-input no-COVID-period feature subset, excluding two period-indicator fields so performance does not rely on COVID-era labels; this subset expands to 229 model columns after one-hot encoding categorical fields. Historical

Table 2 Compact tuning and final-score protocol for gradient-boosted models.

Item	Setting
Feature set	213-input no-COVID-period subset; one-hot expansion to 229 model columns
Tuning sample/seed	Stratified train sample, `max_train_rows` = 300,000, seed 42; selected candidates rebuilt on all 954,990 training rows for final reported scores
LightGBM grid	Depth/leaf/lr: (8,63,0.05), (6,31,0.03), (10,127,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
XGBoost grid	Depth/lr: (8,0.05), (6,0.03), (5,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
Selection	Validation F1; selected LightGBM depth 6, 31 leaves, lr 0.03, sqrt weight; selected XGBoost depth 6, lr 0.03, sqrt weight
Thresholds	Validation grid 0.01–0.99 step 0.005 plus score quantiles; category fallback to global threshold if validation positives < 30

demand features include lag counts, rolling statistics, changes, ratios to historical baselines, current complaint count, and history-availability flags. Calendar features capture seasonality. Weather features summarize weekly temperature, precipitation, snow, wind, and extreme-day counts. OSM POI and PLUTO variables are static neighborhood-context descriptors rather than dynamic covariates; they support predictive association analysis but should not be read as contemporaneous measurements of changing neighborhood conditions.

4.2 Models

The pipeline compares rule baselines, logistic regression, gradient-boosted trees, and tuned LightGBM/XGBoost variants. Numeric preprocessing medians and categorical levels are fitted on the training period only. Categorical fields are one-hot encoded with train-fitted levels, and unseen validation or test categories map to all-zero one-hot indicators for that field. Tuned gradient-boosted models provide the strongest validation performance and capture nonlinear tabular interactions. Hyperparameters and class-weight settings are selected from stratified 300,000-row tuning runs for reproducibility; the final reported LightGBM and XGBoost scores are then rebuilt on all 954,990 target-week training rows using the selected candidates. Scores are converted into hard alerts using thresholds selected only on validation.

4.3 Category-aware ensemble decision layer

Let $s^L_{n,t,c}$ be the tuned LightGBM score and $s^X_{n,t,c}$ the tuned XGBoost score. The ensemble score is

$$s^E_{n,t,c} = 0.5 s^L_{n,t,c} + 0.5 s^X_{n,t,c}.$$

We compare global and complaint-category-specific thresholds. In the category-specific strategy, each category c receives a threshold τ_c selected on the 2023 validation period to optimize F1. The final alert is

$$\hat{y}_{n,t,c} = \mathbf{1}\{s^E_{n,t,c} \geq \tau_c\}.$$

This is category-aware at the decision layer: the same score can imply different alert decisions because score distributions, precision-recall trade-offs, and alert rates differ by service domain.

4.4 Explainability

We use SHAP to explain the tuned LightGBM component. The final decision strategy is an ensemble with category-specific thresholds, so SHAP is interpreted as evidence about a representative tree component trained under the same one-hot feature design and chronological protocol, not as a direct explanation of every ensemble alert or as causal attribution. For reporting, one-hot complaint-category indicators are grouped back into the service-category feature group.

5 Experimental Setup

5.1 Chronological split

The main experiment uses a chronological split by target week, not by feature week: target weeks in 2015-2022 are used for training, 2023 target weeks for validation, and 2024-2025 target weeks for testing. This means, for example, that a row whose feature week ends in late 2022 but predicts an early-2023 target week is assigned to validation rather than training. Rows without an observable next-week target are excluded from model-ready evaluation. This target-week assignment is applied before model fitting, validation ranking, threshold tuning, and test reporting.

The model-ready target-week split contains 954,990 train rows with 121,306 positives, 122,616 validation rows with 15,574 positives, and 247,590 test rows with 31,570 positives. The positive class is rare but stable, with approximately 12.7% positive share in validation and test. This imbalance motivates evaluation with precision, recall, F1, PR-AUC, ROC-AUC, balanced accuracy, and confusion matrices.

5.2 Metrics and model selection

F1 is used as the primary validation-ranking metric because the task requires balancing alert precision and event detection. Precision measures how many alerts correspond to observed positive events, while recall measures how many observed positive events are captured. PR-AUC evaluates ranking quality under class imbalance, and ROC-AUC is reported as a complementary threshold-independent metric. Balanced accuracy is included to summarize sensitivity and specificity without letting the majority class dominate.

All reported thresholds are selected on the validation period only. Threshold tuning is treated as a method component rather than a cosmetic post-processing step because the default 0.5 threshold is poorly matched to an imbalanced early-warning task. Importantly, threshold tuning changes hard classification metrics such as F1, precision, recall, and predicted-positive share; it does not change the underlying PR-AUC or ROC-AUC for a fixed score vector.

Table 3 Final decision-layer comparison. Thresholds are selected on validation and evaluated unchanged on the future test period.

Model	Threshold	Val F1	Val PR-AUC	Test F1	Prec.	Rec.	PR-AUC	ROC-AUC	Bal. acc.	Alert rate
Ensemble LGBM(0.50)+XGB(0.50)	Per category	0.3809	0.3249	0.3839	0.3071	0.5117	0.3365	0.7720	0.6715	0.2124
Tuned XGBoost	Per category	0.3806	0.3233	0.3845	0.3058	0.5175	0.3367	0.7722	0.6729	0.2158
Tuned LightGBM	Per category	0.3802	0.3249	0.3836	0.3022	0.5249	0.3346	0.7709	0.6739	0.2214
Ensemble LGBM(0.50)+XGB(0.50)	Global	0.3777	0.3249	0.3865	0.3001	0.5426	0.3365	0.7720	0.6788	0.2305

6 Results

6.1 Baselines, ablations, and final model comparison

The rule and logistic-regression baselines indicate that simple historical signals alone are insufficient for the final alerting task. On the 2024-2025 target-week test period, current-week persistence obtains F1 = 0.3111 and PR-AUC = 0.2359, a seasonal lag-52 margin obtains F1 = 0.2423 and PR-AUC = 0.1773, and the strongest rule baseline, a rolling z-score rule, obtains F1 = 0.3332 and PR-AUC = 0.2658. Logistic regression with historical features obtains F1 = 0.3256 and PR-AUC = 0.2598. Feature-ablation checks also show that LightGBM ranking quality improves from PR-AUC = 0.2770 with historical features alone to 0.3158 with full context. Final hard-alert performance additionally depends on model tuning, ensemble averaging, and validation-selected thresholds, which are compared in Table 3.

Table 3 compares the final decision strategies. Following the pre-specified validation-selection rule, the validation-selected, operationally motivated category-aware strategy is the LightGBM/XGBoost ensemble with complaint-category-specific thresholds. On the held-out 2024-2025 target-week test period, it achieves F1 = 0.3839, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, ROC-AUC = 0.7720, and balanced accuracy = 0.6715. The global-threshold ensemble has slightly higher test F1, but uses one alert rule across service domains; we retain the validation-selected category-specific strategy for trade-off interpretation.

6.2 Alert interpretation

The final test confusion matrix contains 16,154 true positives, 36,440 false positives, 15,416 false negatives, and 179,580 true negatives. This means the model captures about half of true next-week abnormal reported-demand events while roughly three out of ten alerts correspond to true events. This operating point is more appropriate as a prioritization layer than as an automated allocation rule.

Because the operational use case is prioritization, we also evaluate concentration of true events among the highest-risk predictions. The top 1% of scored test rows contains 2,476 rows with precision 0.5965, recall 0.0468, and 4.68-times lift over the test base rate. The top 5% has precision 0.4548 and recall 0.1784, the top 10% has precision 0.3866 and recall 0.3032, and the full threshold-based alert layer selects 52,594 rows with precision 0.3071, recall 0.5117, and 2.41-times lift.

Compared with logistic-regression and rule baselines, gradient-boosted models improve ranking quality and alert conversion: the rolling z-score rule obtains F1 = 0.3332 and PR-AUC = 0.2658, whereas the selected ensemble obtains F1 = 0.3839 and PR-AUC = 0.3365 under the final validation-selected thresholds.

Table 4 Final model test performance by complaint category on the 2024-2025 target-week period.

Category	Positives	Alerts	PR-AUC	F1	Precision	Recall
Noise	3,982	5,729	0.4229	0.4578	0.3880	0.5583
Public safety	3,536	6,482	0.3312	0.4097	0.3166	0.5803
Housing	3,720	5,412	0.3860	0.4025	0.3396	0.4941
Water/sewer	3,462	6,090	0.3642	0.3938	0.3089	0.5433
Other	3,620	6,695	0.3237	0.3762	0.2898	0.5359
Parking/traffic	3,586	5,811	0.3255	0.3708	0.2998	0.4858
Environment	2,896	5,924	0.3105	0.3633	0.2704	0.5532
Infrastructure	3,596	5,900	0.2764	0.3484	0.2803	0.4600
Sanitation	3,172	4,551	0.2570	0.3165	0.2685	0.3852

6.3 Performance by complaint category

Table 4 shows that performance varies across municipal service domains. Noise has the strongest test F1, followed by public safety, housing, and water/sewer. Sanitation, infrastructure, and environment are weaker by F1, although several retain moderate recall. Category thresholds mainly change the alert conversion trade-off: relative to a global ensemble threshold, the per-category ensemble has macro-F1 0.3821 versus 0.3838, macro precision 0.3069 versus 0.2989, macro recall 0.5107 versus 0.5394, and mean alert rate 0.2124 versus 0.2305. This comparison emphasizes explicit service-domain precision-recall and alert-volume choice rather than aggregate F1 superiority.

7 Explainability Analysis

7.1 Global feature importance

SHAP analysis is performed on the tuned LightGBM component after one-hot categorical encoding. Figure 2 summarizes SHAP importance by feature group, with one-hot complaint-category indicators grouped back into the service-category group. Historical temporal features dominate, accounting for 58.96% of summed mean absolute SHAP importance. This pattern is consistent with the target definition, which is based on deviations from recent local history: recent volatility, ratios to historical means, seasonal recurrence, and lag counts directly encode demand dynamics. However, contextual features still carry measurable signal. Service category accounts for 10.73%, PLUTO built environment for 10.18%, OSM POI context for 8.13%, weather for 4.88%, calendar features for 4.77%, and current demand for 1.61%.

The leading individual features are consistent with the next-week abnormal reported-demand formulation: `rolling_8w_std`, `ratio_to_8w_mean`, `week_of_year`, `lag_52w_count`, `rolling_4w_mean`, `lag_1w_count`, service-category indicators such as `complaint_category=environment`, and current `complaint_count`. Together they capture volatility, baseline deviation, seasonality, service domain, and current demand.

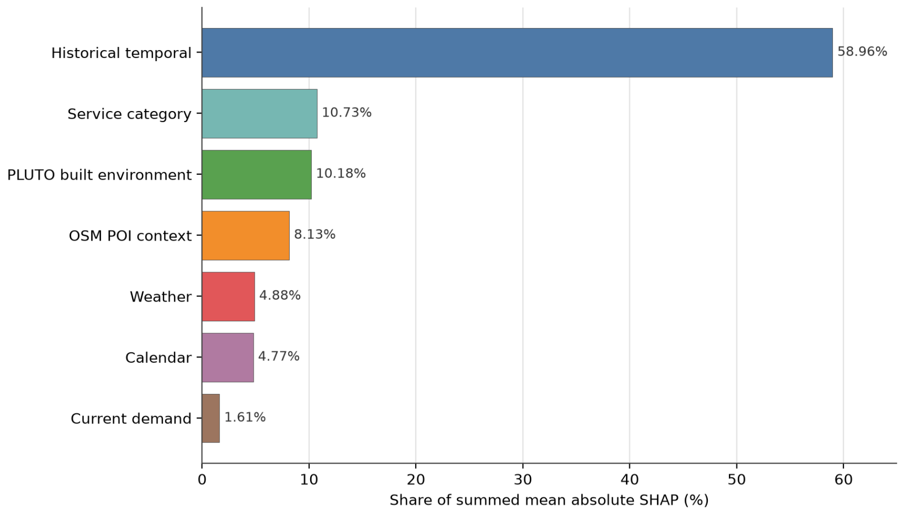

Fig. 2 Normalized summed mean absolute SHAP importance for the tuned LightGBM explanation backbone. Historical temporal demand features dominate, while service-category, PLUTO, OSM POI, weather, and calendar features add smaller but interpretable contextual contributions.

7.2 Category-specific interpretation

Across service categories, the highest-ranked SHAP features are broadly similar but differ in relative magnitude. Top features consistently include rolling eight-week standard deviation, ratio to the eight-week mean, complaint-category indicators, 52-week lag count, and week of year. This pattern supports two interpretations. First, the model is primarily learning temporal abnormality relative to local historical behavior. Second, service category remains important even after category-specific series are included, suggesting that service domains differ in baseline volatility, seasonality, and alertability.

Two local SHAP examples further show how the explanation layer should be used. A true-positive housing alert in Inwood for feature week 2025-10-06 has a LightGBM score of 0.954 and is associated with eight-week volatility, seasonal timing, and residential built-environment features. A false-positive public-safety alert in the Lower East Side for feature week 2025-06-30 has a LightGBM score of 0.940; annual and recent lag counts and current reported demand push the score upward, while eight-week volatility and zero weekly precipitation push it downward. These examples illustrate why explanations should support analyst review rather than automatic action.

The built-environment and POI contributions should be interpreted as contextual associations. PLUTO density or residential-area features may help distinguish neighborhoods with different housing-related demand dynamics, while POI entropy and named POI counts may reflect functional intensity or mixed-use conditions. These explanations do not prove that changing a POI or land-use feature would causally change 311 demand; they describe how the trained model uses observed context to estimate reported-demand risk.

8 Discussion

8.1 Operational meaning

The final model is best understood as early-warning decision support. On the future test period, recall of 0.5117 means it captures about half of true next-week abnormal reported-demand events, while precision of 0.3071 means many alerts are false positives. In practice, the system should be used as a triage layer that narrows analyst attention to high-risk NTA-category weeks, with final decisions incorporating agency capacity, complaint severity, and local knowledge.

Complaint-category-specific thresholds make this interpretation more practical. A single global threshold assumes all domains share the same alert rule, but noise, housing, infrastructure, sanitation, and public-safety requests differ in baseline frequency, volatility, reporting behavior, and response costs. A category-aware threshold layer reflects these differences while retaining a common modeling framework. The observed F1 difference between global and category-specific thresholds is small, so the main claim is explicit precision-recall and alert-volume trade-off by service domain rather than aggregate performance superiority.

8.2 Methodological implications

The experiment highlights three practical issues for this type of forecasting. Temporal leakage control is essential because random splits are poorly aligned with prospective prediction in repeated neighborhood-category panels. Threshold selection also needs to be reported explicitly in imbalanced early-warning tasks, since the default 0.5 threshold can produce impractical alert rates. Context features are most useful as complements to historical demand: SHAP shows that temporal dynamics dominate, but neighborhood function, built environment, weather, service category, and calendar timing still shape event risk.

8.3 Limitations

Several limitations should guide interpretation. NYC 311 data measure reported service demand, not all underlying urban problems, and reporting propensity may reproduce socio-spatial inequities. OSM POI and PLUTO are static context descriptors with snapshot/look-ahead risk across 2015-2025, and NOAA weather is weekly city-level exposure rather than NTA microclimate. Hyperparameters are selected from sampled tuning runs, although final reported scores are rebuilt on all 954,990 training rows; three sampled seeds produced test F1 0.3817 ± 0.0021, PR-AUC 0.3313 ± 0.0025, and ROC-AUC 0.7678 ± 0.0007. Future work should examine whether the findings hold under alternative abnormal-demand definitions, in other cities, across repeated full-training runs, and when linked to operational response outcomes.

9 Conclusion

This paper presented an explainable machine-learning pipeline for forecasting next-week abnormal reported demand in NYC 311 data at the neighborhood, week, and service-category level. The NTA-week-complaint-category formulation, multi-source context, and target-week chronological split frame 311 forecasting as an early-warning task rather than generic count prediction. The validation-selected LightGBM/XGBoost ensemble with complaint-category-specific thresholds achieved future-test F1 = 0.3839, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, and ROC-AUC = 0.7720. SHAP showed that historical temporal dynamics dominate, while built environment, POI context, weather, service category, and calendar features add smaller but interpretable contextual contributions. The findings support category-aware decision layers as a way to make service-domain alert trade-offs explicit, while aggregate F1 remains similar to a global-threshold policy; future work should test transferability, fairness, alternative abnormal-demand definitions, and operational response links.

Declarations

Data and code availability. Source data are NYC Open Data 311 (`erm2-nwe9`), NOAA GHCN-Daily station `GHCND:USW00094728`, OpenStreetMap Overpass, NYC PLUTO/MapPLUTO (`64uk-42ks`), and 2020 NTA boundaries (`9nt8-h7nd`); the local snapshot was extracted on 2026-06-25. Scripts, mappings, configurations, and evaluation code are available at https://github.com/PhongLam26/SIMC_NYC; processed data release is subject to licensing, redistribution, and storage constraints.

Competing interests. The authors declare no competing interests.

Funding. No specific funding was received.

Author contributions. Tran Dai Phong Lam led data processing, modeling, analysis, and drafting; Thu Le served as academic advisor and supervised the study design, methodology, and manuscript editing; Nguyen Quoc Hung and Nguyen Trung Trinh contributed validation and review.

References

[1] Wang, L., Qian, C., Kats, P., Kontokosta, C.E., Sobolevsky, S.: Structure of 311 service requests as a signature of urban location. PLOS ONE **12**(10), e0186314 (2017) https://doi.org/10.1371/journal.pone.0186314

[2] Kontokosta, C.E., Hong, B.: Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. Sustainable Cities and Society **64**, 102503 (2021) https://doi.org/10.1016/j.scs.2020.102503

[3] Zheng, Y., Capra, L., Wolfson, O., Yang, H.: Urban computing: Concepts, methodologies, and applications. ACM Transactions on Intelligent Systems and Technology **5**(3), Article 38, 1–55 (2014) https://doi.org/10.1145/2629592

[4] Agonafir, C., Ramirez Pabon, A., Lakhankar, T., Khanbilvardi, R., Devineni, N.: Understanding New York City street flooding through 311 complaints. Journal of Hydrology **605**, 127300 (2022) https://doi.org/10.1016/j.jhydrol.2021.127300

[5] Haklay, M., Weber, P.: OpenStreetMap: User-generated street maps. IEEE Pervasive Computing **7**(4), 12–18 (2008) https://doi.org/10.1109/MPRV.2008.80

[6] Zhang, L., Pfoser, D.: Using OpenStreetMap point-of-interest data to model urban change—a feasibility study. PLOS ONE **14**(2), e0212606 (2019) https://doi.org/10.1371/journal.pone.0212606

[7] Saito, T., Rehmsmeier, M.: The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE **10**(3), e0118432 (2015) https://doi.org/10.1371/journal.pone.0118432

[8] Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. ACM, New York, NY, USA (2016). https://doi.org/10.1145/2939672.2939785

[9] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y.: LightGBM: A highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems, vol. 30 (2017). https://papers.nips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree

[10] Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems, vol. 30 (2017). https://arxiv.org/abs/1705.07874

[11] Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.-I.: From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence **2**, 56–67 (2020) https://doi.org/10.1038/s42256-019-0138-9

[12] Rudin, C.: Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence **1**, 206–215 (2019) https://doi.org/10.1038/s42256-019-0048-x